

HASSAN AHMED

Full Stack Developer | Next.js | NestJS | Node.js | PostgreSQL

Islamabad, Pakistan | +92 342 0042947 | hassan87691@gmail.com

linkedin.com/in/hassan-ahmed-1595b12a6 | github.com/hassan2314 | Portfolio

SUMMARY

Full Stack Developer building production platforms serving 50,000+ users with Next.js, NestJS, and PostgreSQL. Strong in REST API design, secure authentication and role-based access, database modeling and query optimization, AI service integration, and building admin panels and CMS tooling used daily by non-technical teams.

PROFESSIONAL EXPERIENCE

Full Stack Developer — Digital Matters

Sep 2025 – Present

ReadMaududi — Digital Library Platform (50,000+ readers) | readmaududi.com

- Build and own the backend (NestJS, PostgreSQL, Prisma, JWT auth) and Next.js admin panel of a digital library serving 50,000+ readers, with REST APIs and full content management for books, authors, categories, and publishers.
- Developed content ingestion and OCR pipelines (Google Document AI, AWS S3) to digitize and publish book content, and implemented in-app subscriptions with Apple App Store and Google Play billing.
- Integrated external AI services into production APIs and admin tools to power AI chat, search, and summarization features.
- Optimized browsing with server-side pagination, multi-criteria filtering, and database indexing on frequently queried columns.

Official Web Platform — National Non-Profit Organization

Building the official bilingual (English/Urdu with RTL support) web platform of a large national non-profit organization using Next.js 15, React 19, NestJS, Sequelize, and PostgreSQL.

- Architected a modular backend of 25+ NestJS modules — custom CMS with post revisions, dynamic page builder with configurable layouts and sections, events, campaigns, donations, and membership registration — secured with JWT authentication and role-based access control.
- Implemented a media pipeline with AWS S3 presigned uploads and on-the-fly image optimization (Sharp), plus response caching and compression to keep the platform fast under high traffic.
- Built interactive frontend features including a d3-geo interactive regional map, rich-text content editing, localized routing, and SEO with dynamic sitemaps and metadata, using React Query and Tailwind CSS.
- Containerized services with Docker and manage production deployments with PM2; documented all APIs with Swagger.

Full Stack Intern — Red Buffer

Jun 2025 – Aug 2025

- Built an AI text summarization tool (React, Node.js, FastAPI) integrating Hugging Face transformer models with Gemini and Groq LLM endpoints, secured with JWT authentication and real-time response handling.

Web Developer Intern — Elites Digital Marketing

Sep 2024 – Feb 2025

- Developed responsive landing pages and marketing microsites (HTML, CSS, JavaScript) with SEO-driven UI improvements, and handled production deployments.

Software Developer Intern — Pakistan Post

Jun 2023 – Aug 2023

- Contributed to digitizing branch operations (parcel records, staff screens) with React, Node.js, and MongoDB, implementing CRUD workflows with authentication and role-based access.

PROJECTS

Stitch & Style — 3D Custom Fabric Design Studio (Final Year Project)

GitHub

3D clothing customization platform with real-time previews, dynamic fabric selection, designer commission calculation, order placement, and Three.js-based visualization. Tech: React, Node.js, MongoDB, Three.js

Next Video — Video Sharing Platform

Live

Video-sharing platform with channel-based video management, NextAuth session-based access control, and optimized media upload and delivery via ImageKit. Tech: Next.js 15, React 19, TypeScript, MongoDB, NextAuth.js, ImageKit

Acme Docs Copilot — Agentic RAG Knowledge Assistant

Live | API Docs

Internal knowledge assistant that grounds every answer in source documents via a multi-agent architecture (Google ADK) — a coordinator routes to a QA agent (search-then-answer, refuses to guess) or a summarization agent (full-document read). Deterministic ingest pipeline (chunk → Gemini embeddings → pgvector) is kept separate from the agentic query path; every response returns citations and an agent trace. Tech: React, FastAPI, Google ADK, Gemini, PostgreSQL/pgvector — deployed on Netlify, FastAPI Cloud, and Railway.

SKILLS

Languages: JavaScript, TypeScript, SQL, HTML, CSS

Frontend: Next.js, React, Tailwind CSS, React Query (TanStack), Framer Motion

Backend: NestJS, Node.js, Express, REST APIs, JWT & Role-Based Access, Swagger

Databases & ORMs: PostgreSQL, MongoDB, Prisma, Sequelize, Mongoose

AI Integration: LLM APIs (Gemini, Groq), Google ADK (multi-agent orchestration), pgvector, Google Document AI (OCR), Hugging Face

Tools & DevOps: Git, Docker, AWS S3, PM2, Vercel, Postman

Spoken Languages: English, Urdu

EDUCATION

Bachelor of Science in Computer Science — National University of Modern Languages (NUML)

2021 – 2025

Islamabad